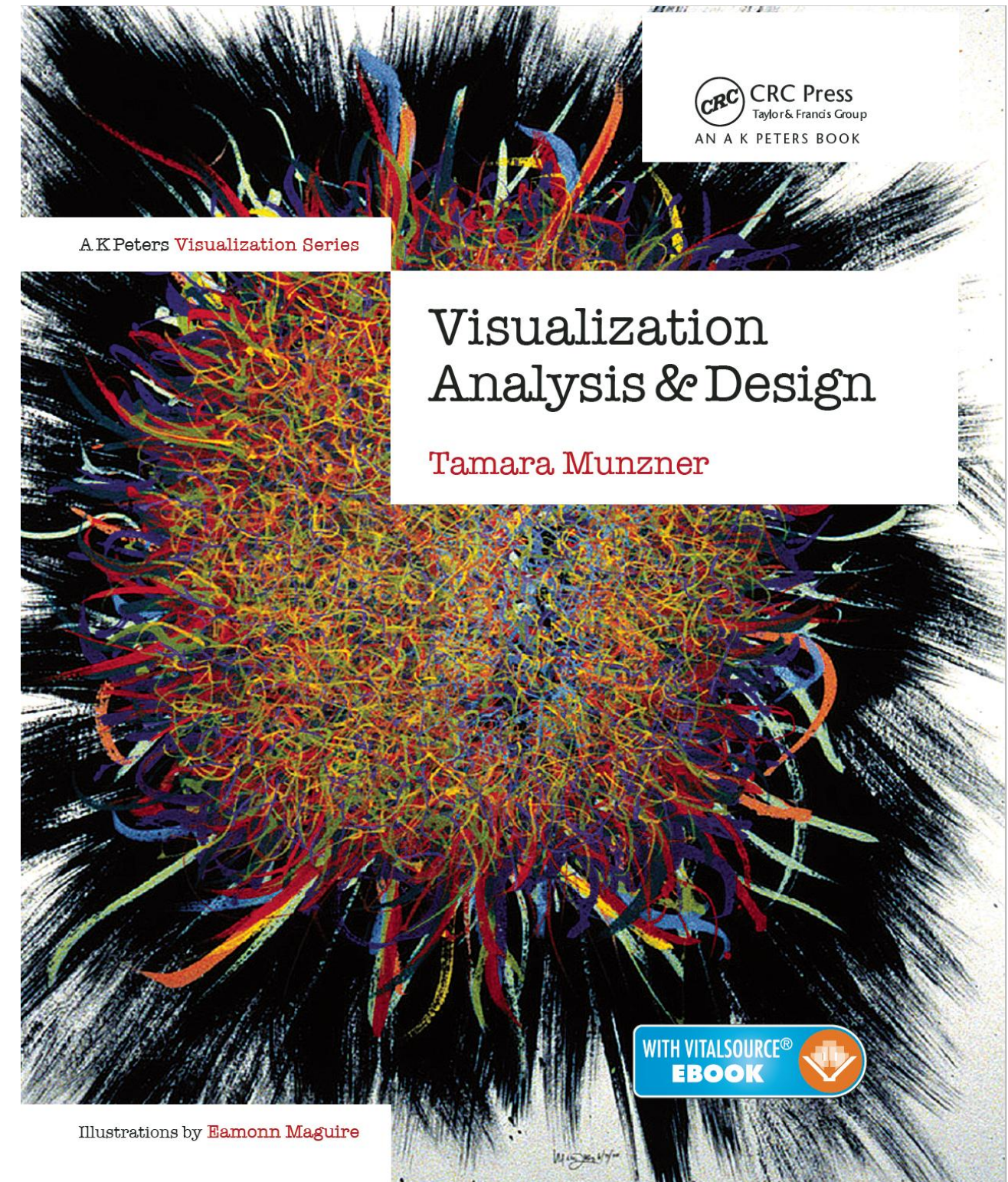


CDS505

Data Visualization & Visual Analytics

Rules of Thumb

*Nur Intan Raihana Ruhaiyem, PhD.
School of Computer Sciences
Universiti Sains Malaysia*

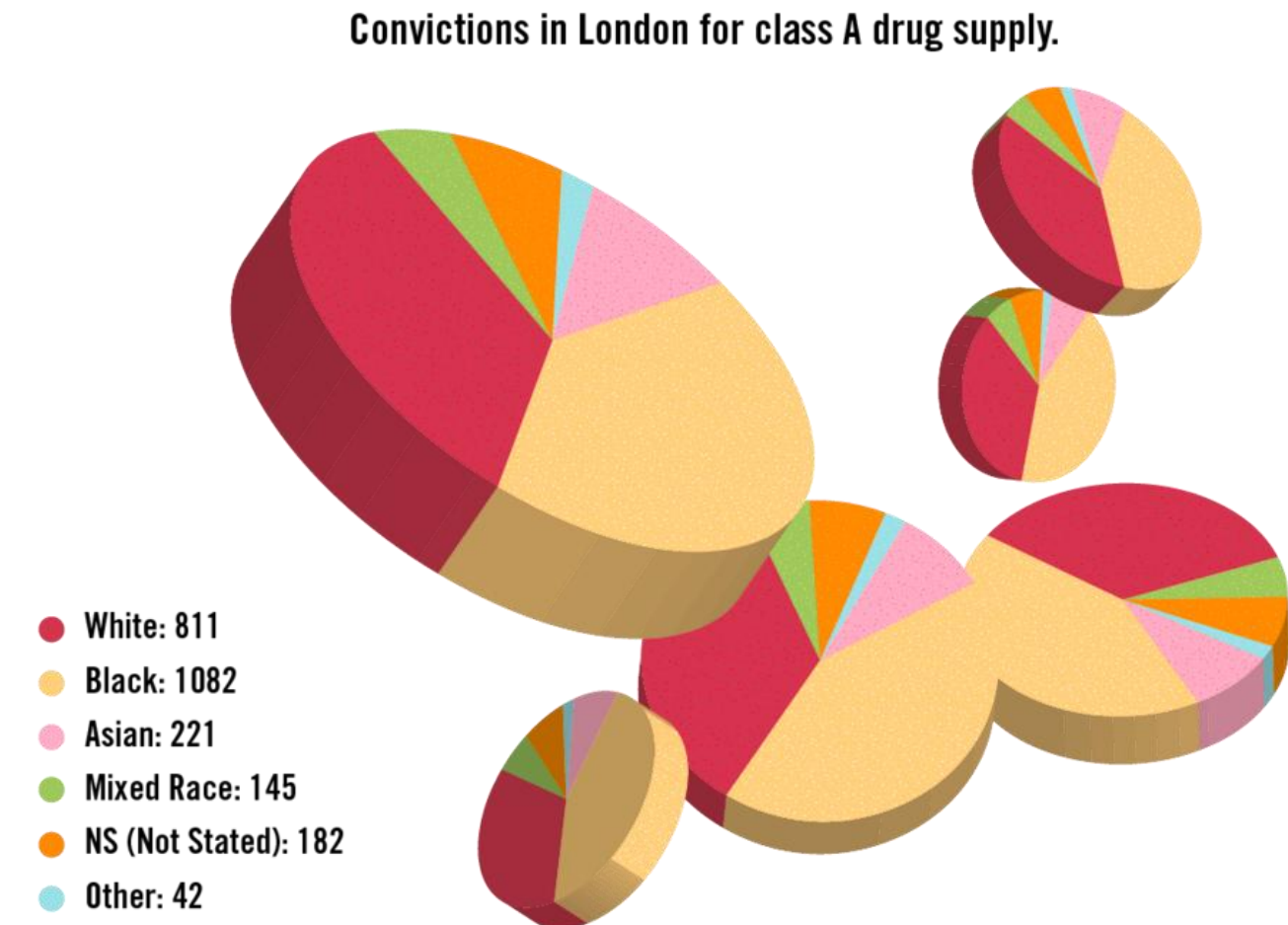
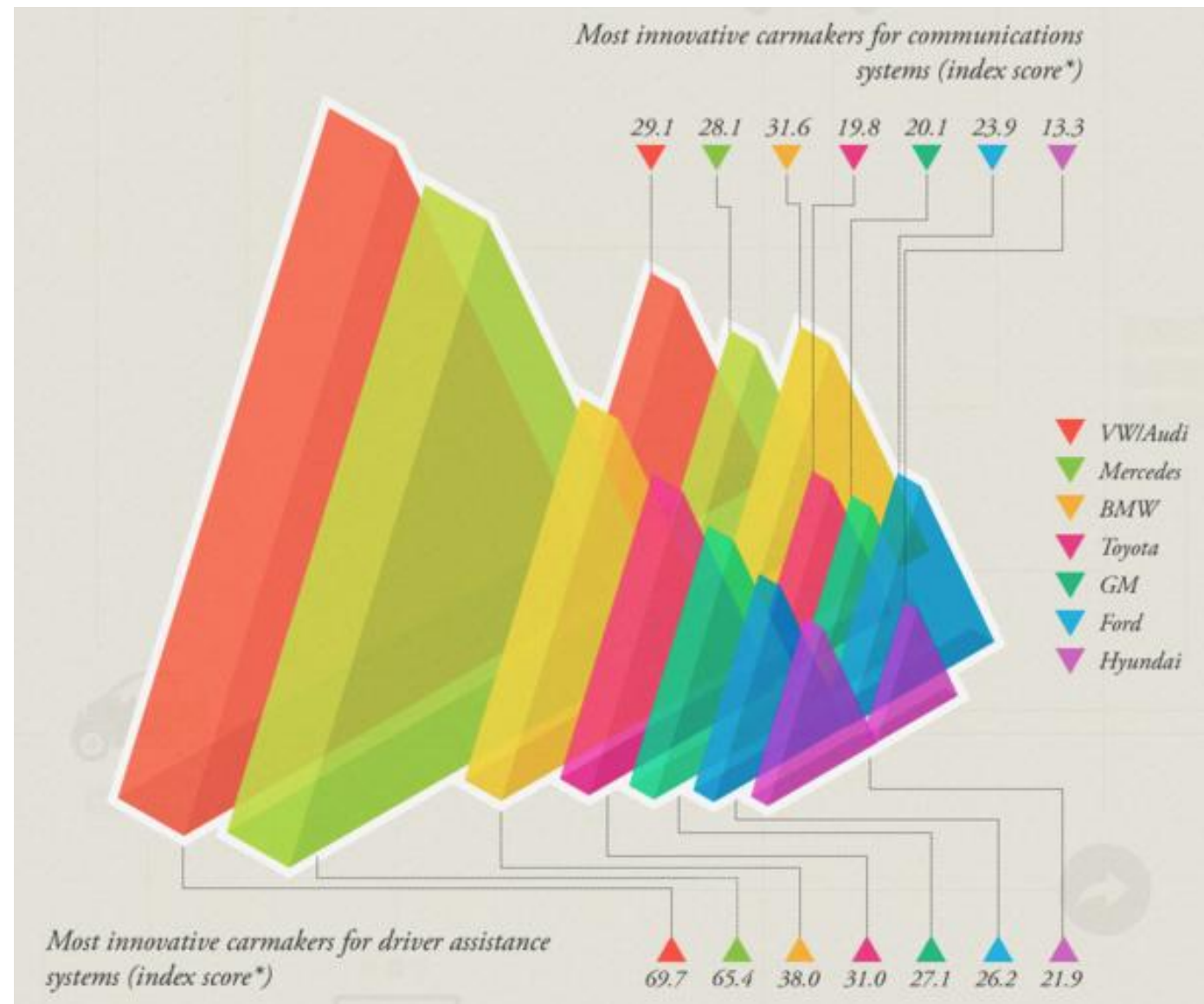


- ✓ Why and When to Follow Rules of Thumb?
- ✓ Eight Rules of Thumbs:
 1. No unjustified 3D
 2. No unjustified 2D
 3. Eyes beat memory
 4. Resolution over immersion
 5. Overview first, zoom and filter, details on demand
 6. Response & interaction
 7. Get it right in black and white
 8. Function first, form next

Why and When to Follow Rules of Thumb?

- Based on empirical studies, own experience, and some have been proposed in previous work.
- They are not set in stone; indeed, they are deeply incomplete. **WHY?**
- The characterization of what idioms are appropriate for which task and data abstractions is still an on-going research frontier.
- There are many open questions.

1. No Unjustified 3D



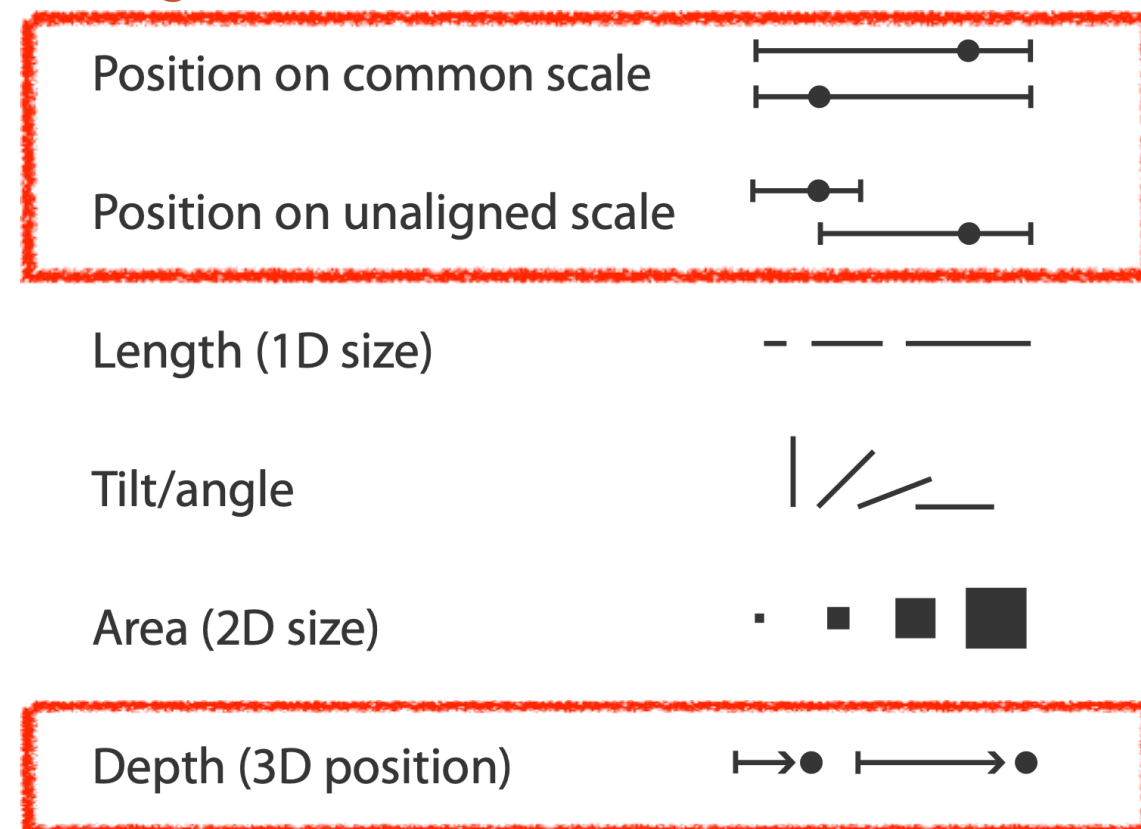
<http://viz.wtf/post/137826497077/eye-popping-3d-triangles>

<http://viz.wtf/post/139002022202/designer-drugs-ht-ducqn>

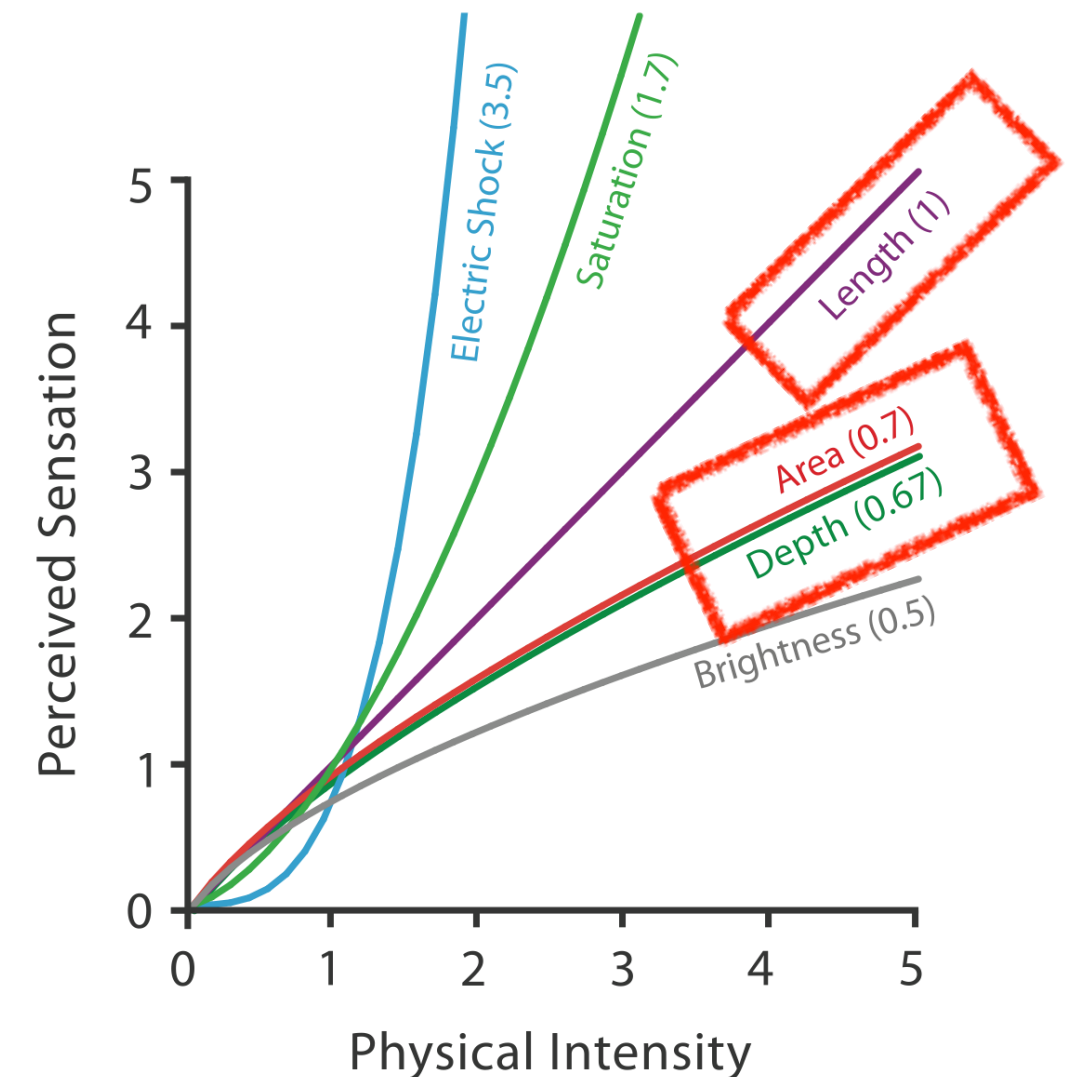
Depth vs power of the plane

- high-ranked spatial position channels: **planar** spatial position
 - not depth!

➔ Magnitude Channels: Ordered Attributes



Steven's Psychophysical Power Law: $S = I^N$



1. No Unjustified 3D – (b) Disparity/Danger of Depth

We don't really live in 3D → we **see** in 2.05D

- acquire more info on image plane quickly from eye movements
- acquire more info for depth slower, from head/body motion

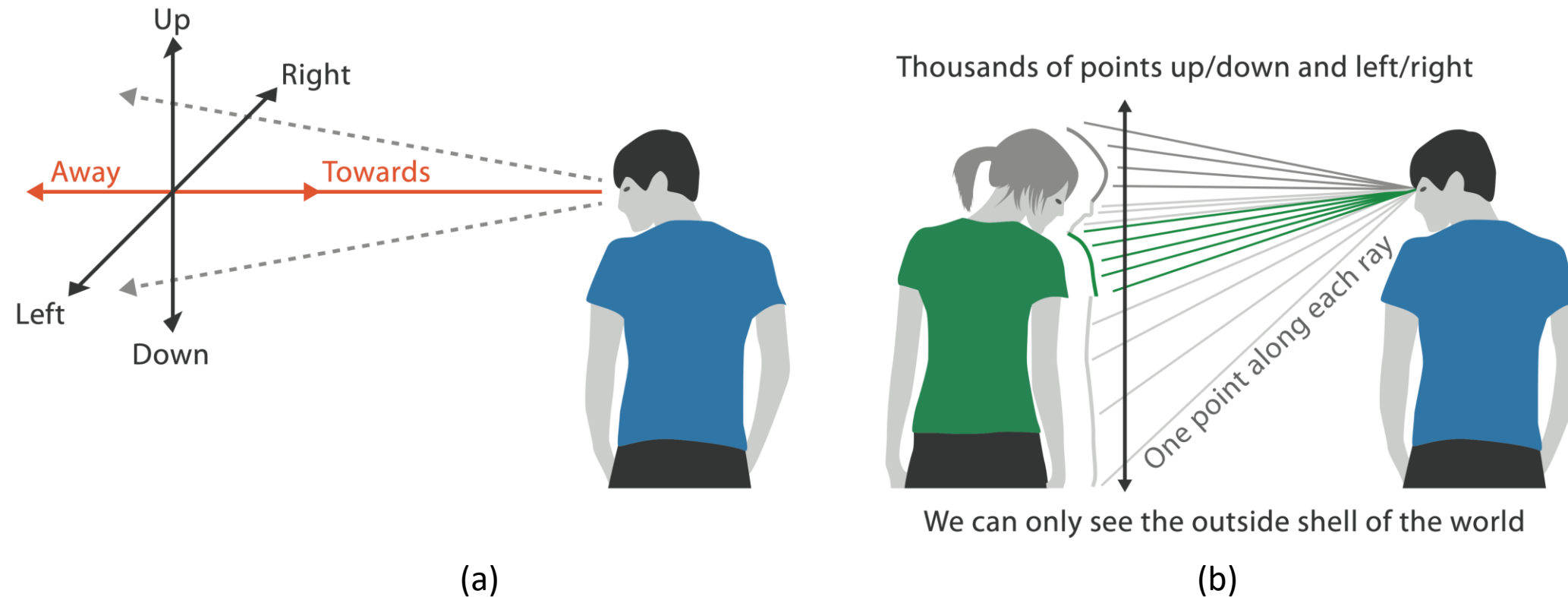
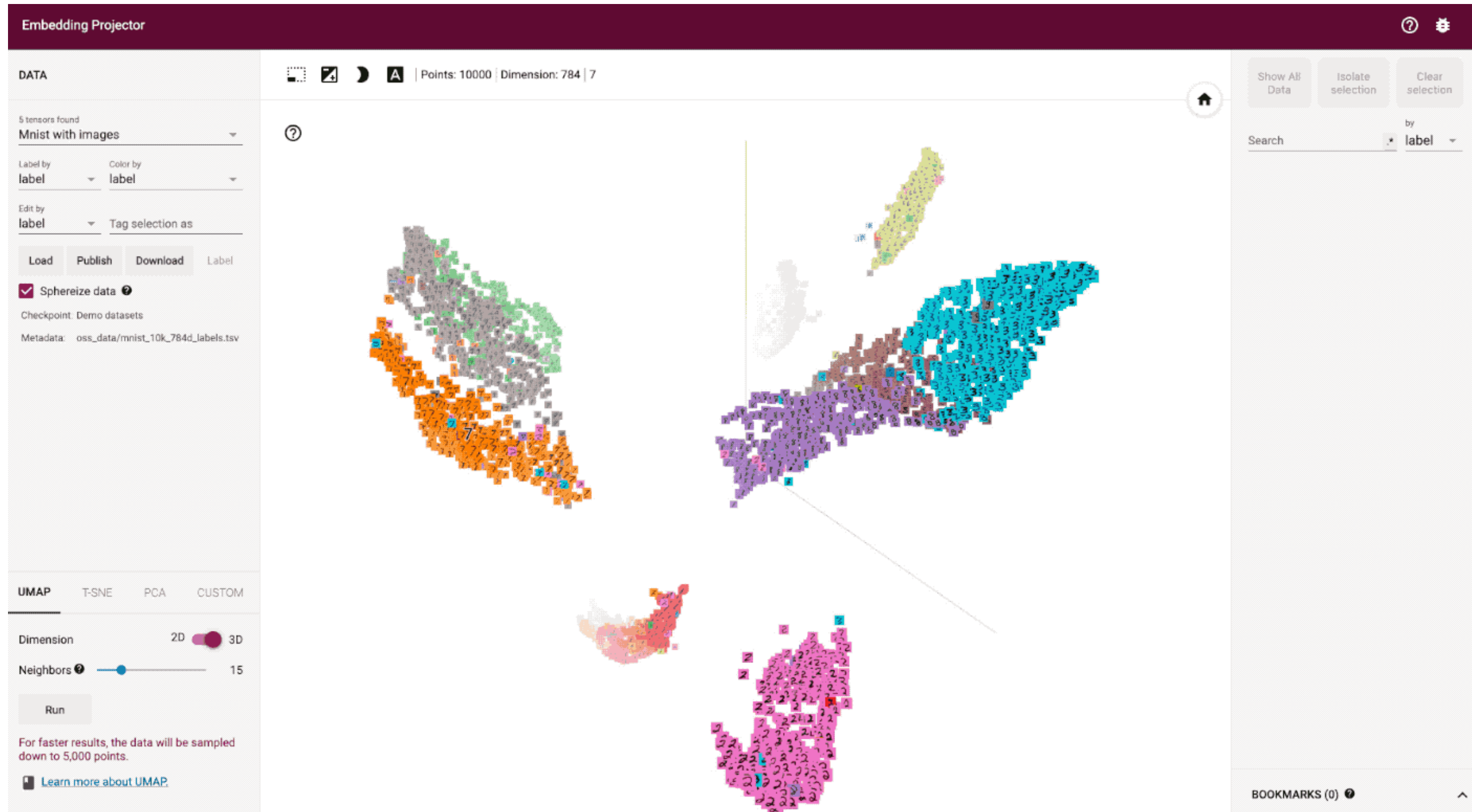


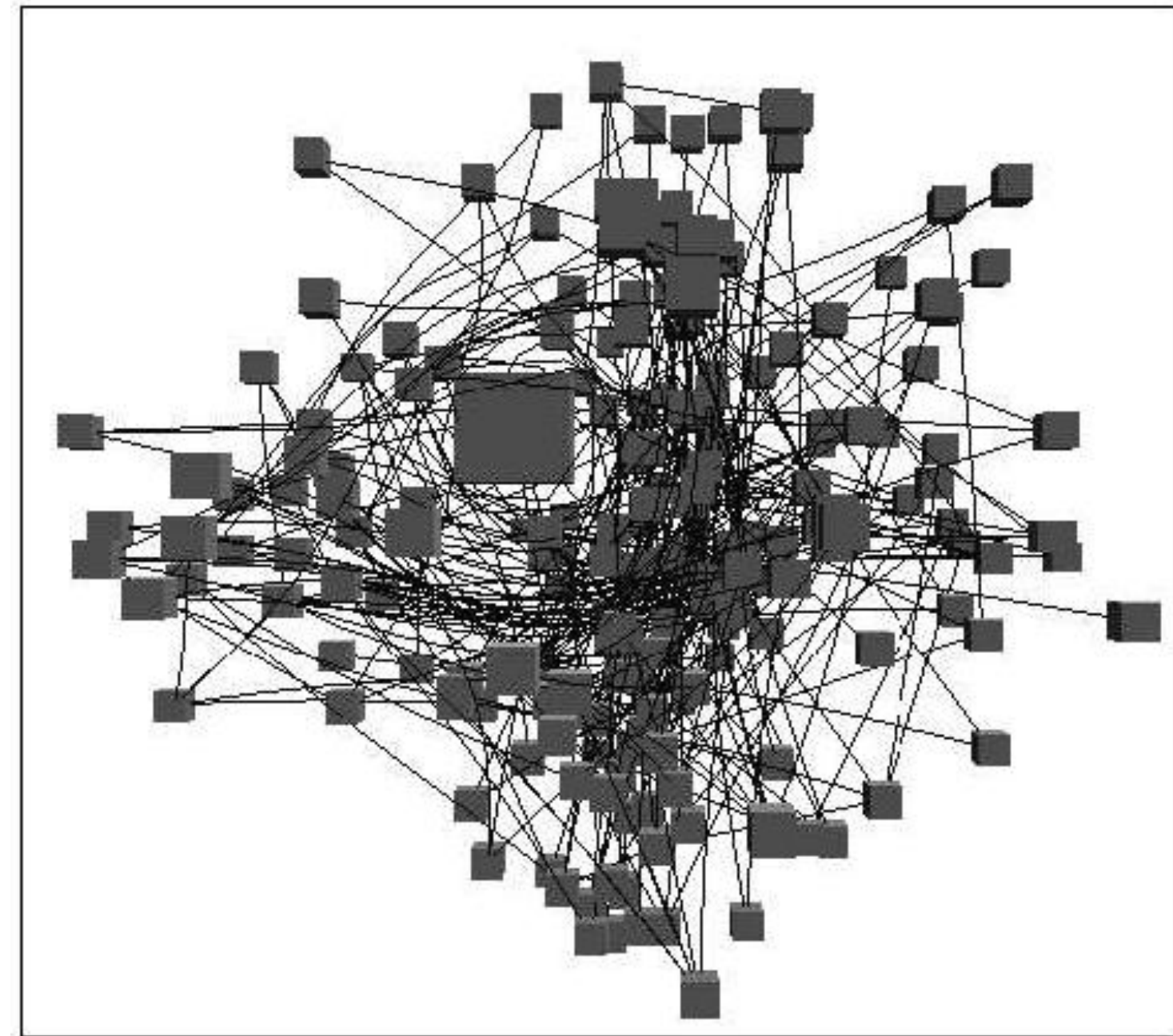
Figure 6.2. Seeing planar position versus depth. (a) The sideways and up-down axes are fundamentally different from the toward-away depth axis. (b) Along the depth axis we can see only one point for each ray, as opposed to millions of rays for the other two axes. After [Ware 08, page 44]

1. No Unjustified 3D – (b) Disparity/Danger of Depth



1. No Unjustified 3D – (c) Occlusion Hides Information

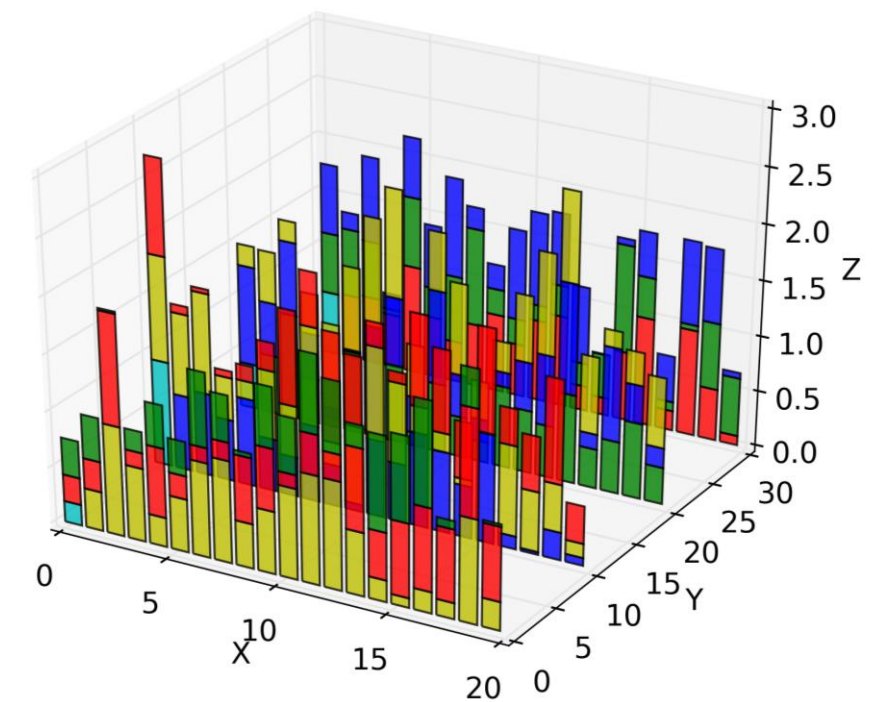
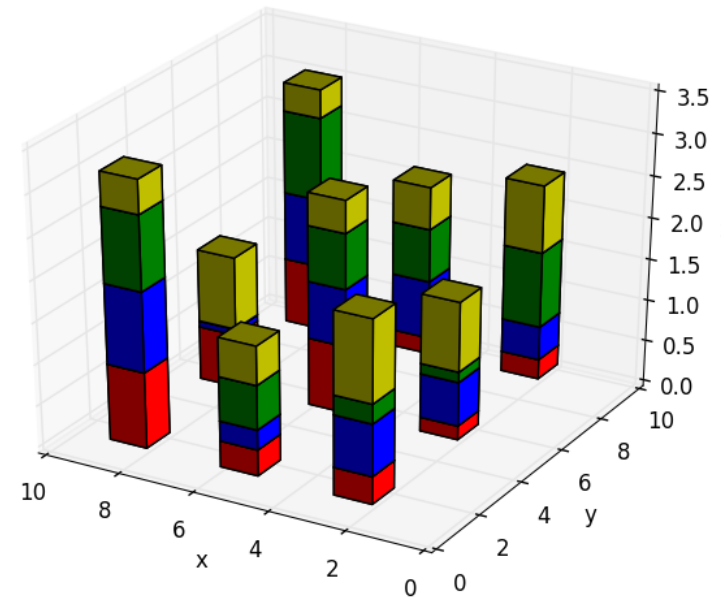
- **Occlusion** - some objects cannot be seen because they are hidden.
- The occlusion relationships between objects change as we move around; this **motion parallax** allows us to build up an understanding of the relative distances between objects.
- **Interaction** complexity - takes time and imposes cognitive load, even with sophisticated interaction idioms.



[Distortion Viewing Techniques for 3D Data. Carpendale et al. InfoVis1996.]

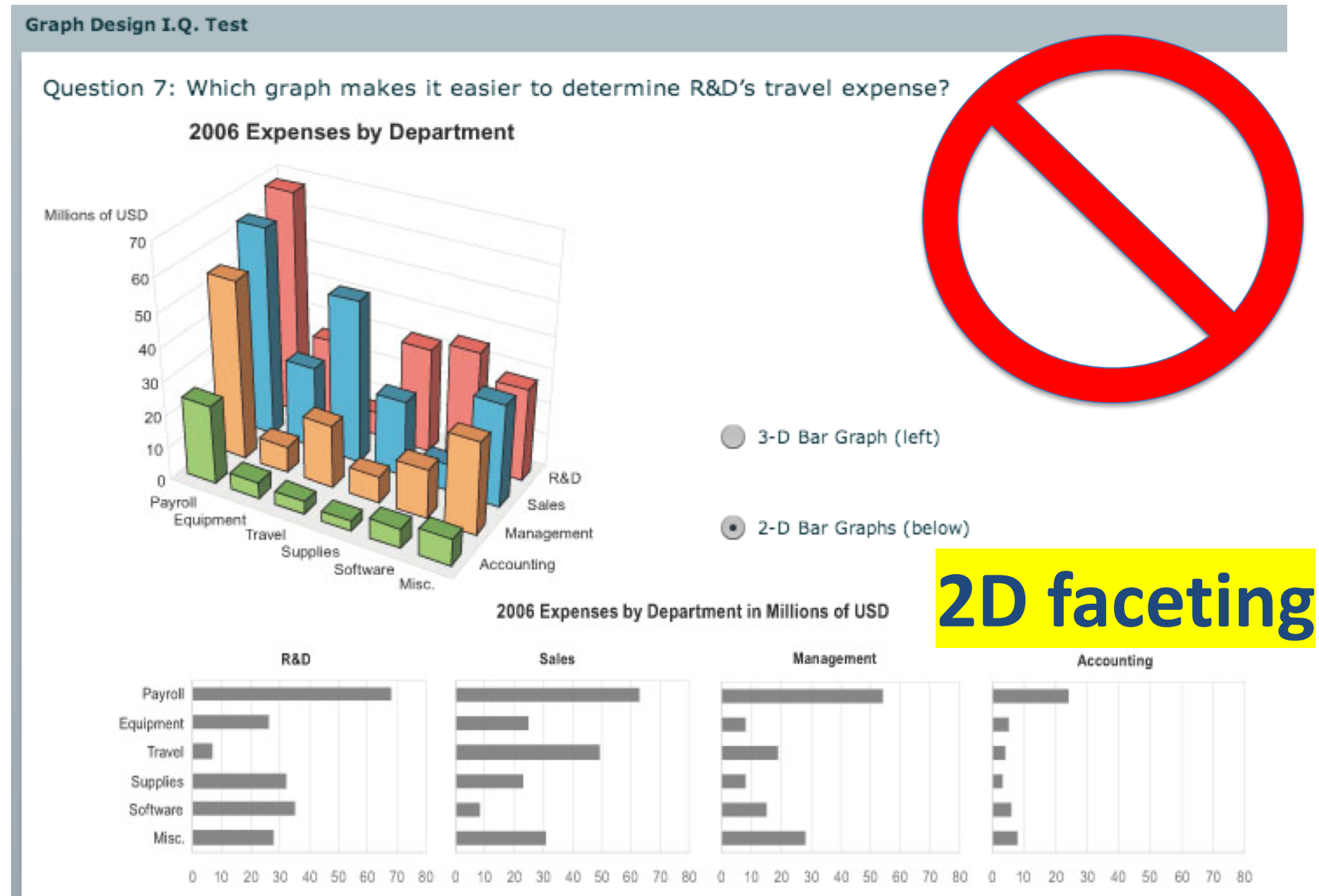
1. No Unjustified 3D – (d) Perspective Distortion Loses Information

- **Perspective distortion** - distant objects appear smaller and change their planar position on the image plane. Also called **foreshortening**.
- 3D vs 2D bar charts
 - ✓ 3D bars **never** a good idea!
 - ✓ Both perspective distortion and occlusion.



1. No Unjustified 3D – (d) Perspective Distortion Loses Information

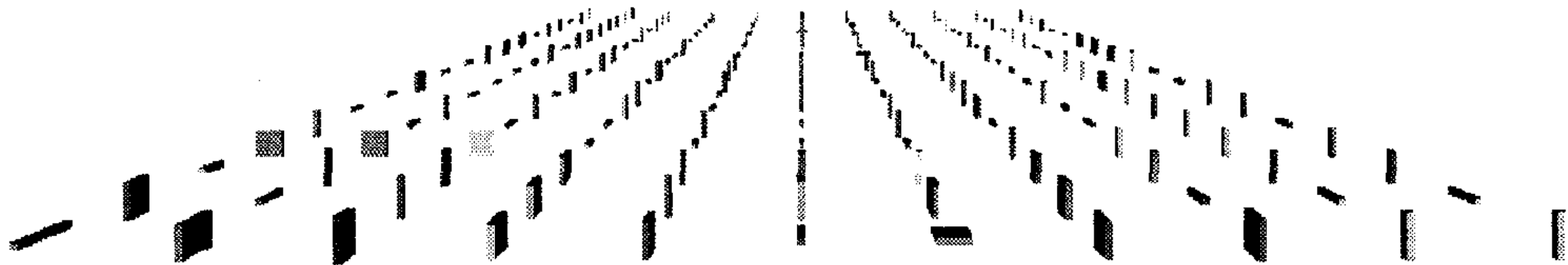
- **Perspective distortion** - distant objects appear smaller and change their planar position on the image plane. Also called **foreshortening**.
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<http://perceptualedge.com/files/GraphDesignIQ.html>

Perspective distortion

- interferes with all **size channel** encodings
- **power of the plane** is lost!



[Visualizing the Results of Multimedia Web Search Engines. Mukherjea, Hirata, and Hara. InfoVis 96]

- The depth cues of shadows and surface shading also communicate depth and three-dimensional structure information.
- Cast shadows are useful for resolving depth ambiguity because they allow us to infer the height of an object with respect to a ground plane.
- Problems - they create visual clutter, could be mistaken by the viewer for true marks, occluding of true marks and surface shading effects interfere with the colour channels: highlights can change the hue or saturation, and shadows change the luminance.

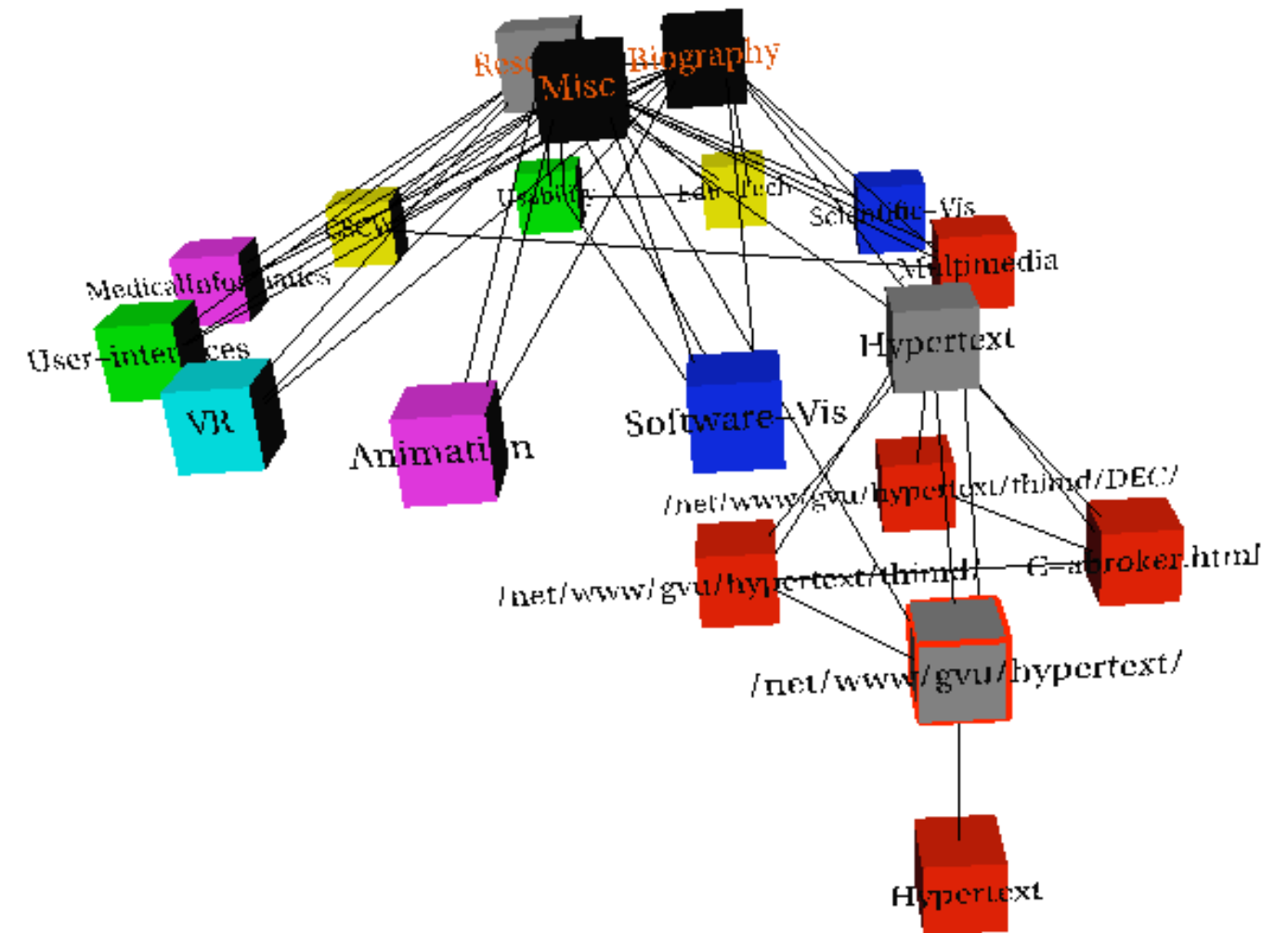
1. No Unjustified 3D – (e) Other Depth Cues – Stereoscopic Depth

- A cue that comes from the disparities between two images made from two camera viewpoints slightly separated in space.
- Although many people assume that stereo vision is the strongest depth cue, it is in fact a relatively weak one compared with the others listed above and contributes little for distant objects.
- Stereo depth cues are most useful for nearby objects, within arm's reach.
- Stereo displays do help people better resolve depth. - but even still depth cannot be perceived with the accuracy of planar position.

1. No Unjustified 3D – (f) Tilted Text Isn't Legible

- Text fonts have been very carefully designed for maximum legibility when rendered on the grid of pixels that makes up a 2D display
- Text legibility
 - far **worse** when tilted from the image plane
- Further reading:

[Exploring and Reducing the Effects of Orientation on Text Readability in Volumetric Displays. Grossman et al. CHI 2007]make

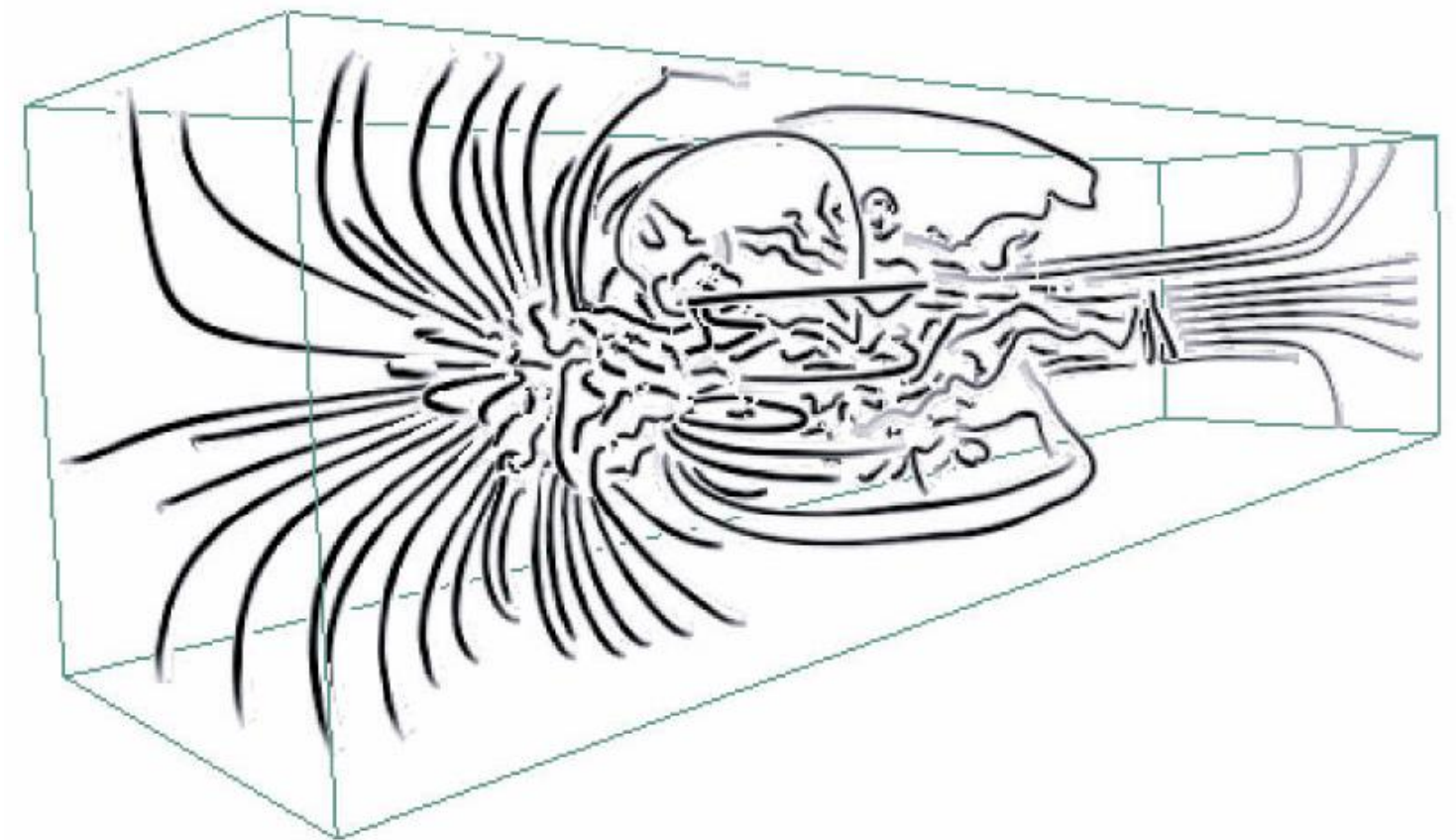


[[Visualizing the World-Wide Web with the Navigational View Builder](#). Mukherjea and Foley. Computer Networks and ISDN Systems, 1995.]]

1. No Unjustified 3D – (g) Justified 3D – Shape Perception

- Benefits outweigh costs when task is shape perception for 3D spatial data
 - interactive navigation supports synthesis across many viewpoints

3D streamline showing the patterns of fluid flow through a volume.

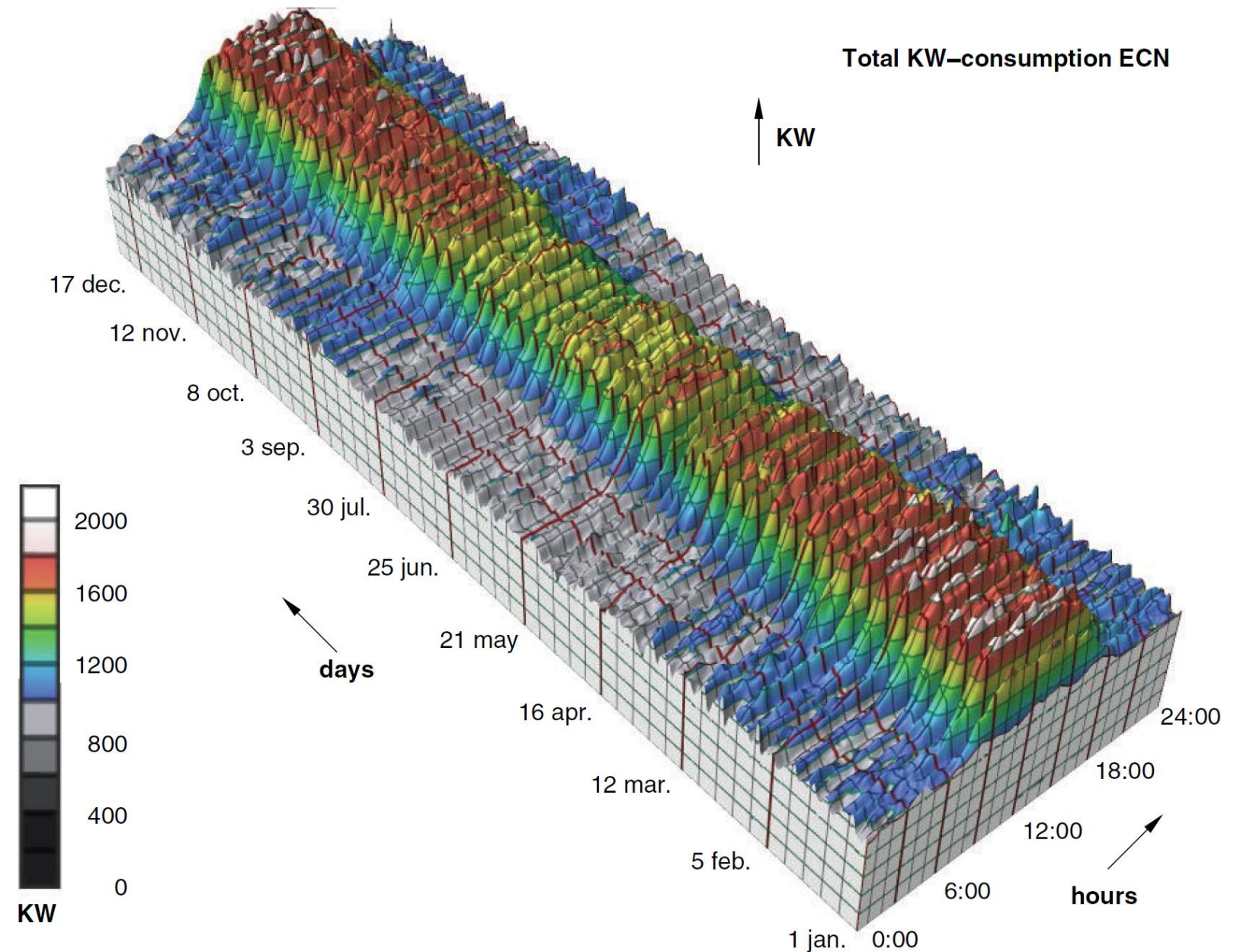


[Image-Based Streamline Generation and Rendering. Li and Shen. IEEE Trans. Visualization and Computer Graphics (TVCG) 13:3 (2007), 630–640.]

1. No Unjustified 3D Example – Time-series Data

Extruded curves:

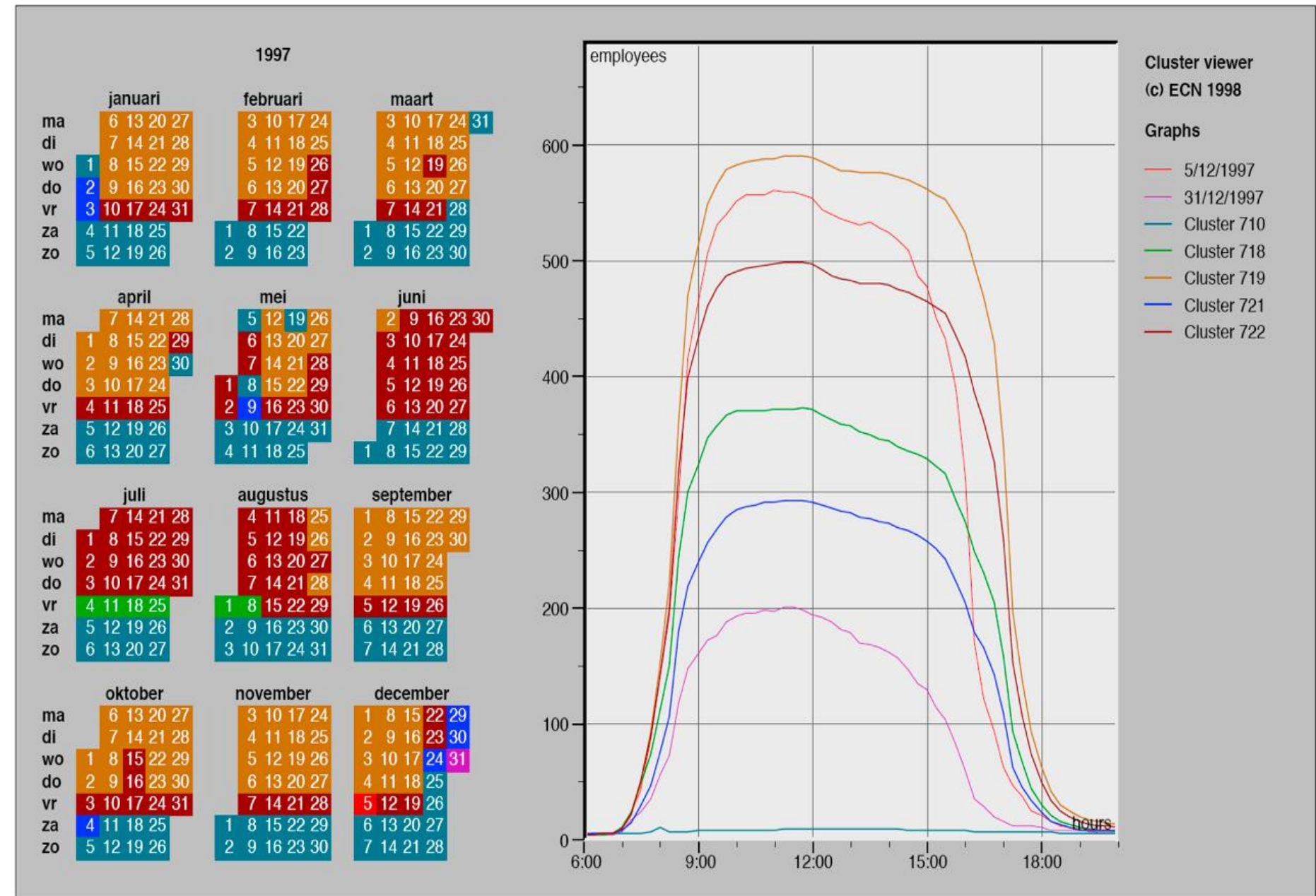
detailed comparisons
impossible, occlusion and
perspective distortion



[\[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.\]](#)

1. No Unjustified 3D Example – Transform for New Data Abstraction

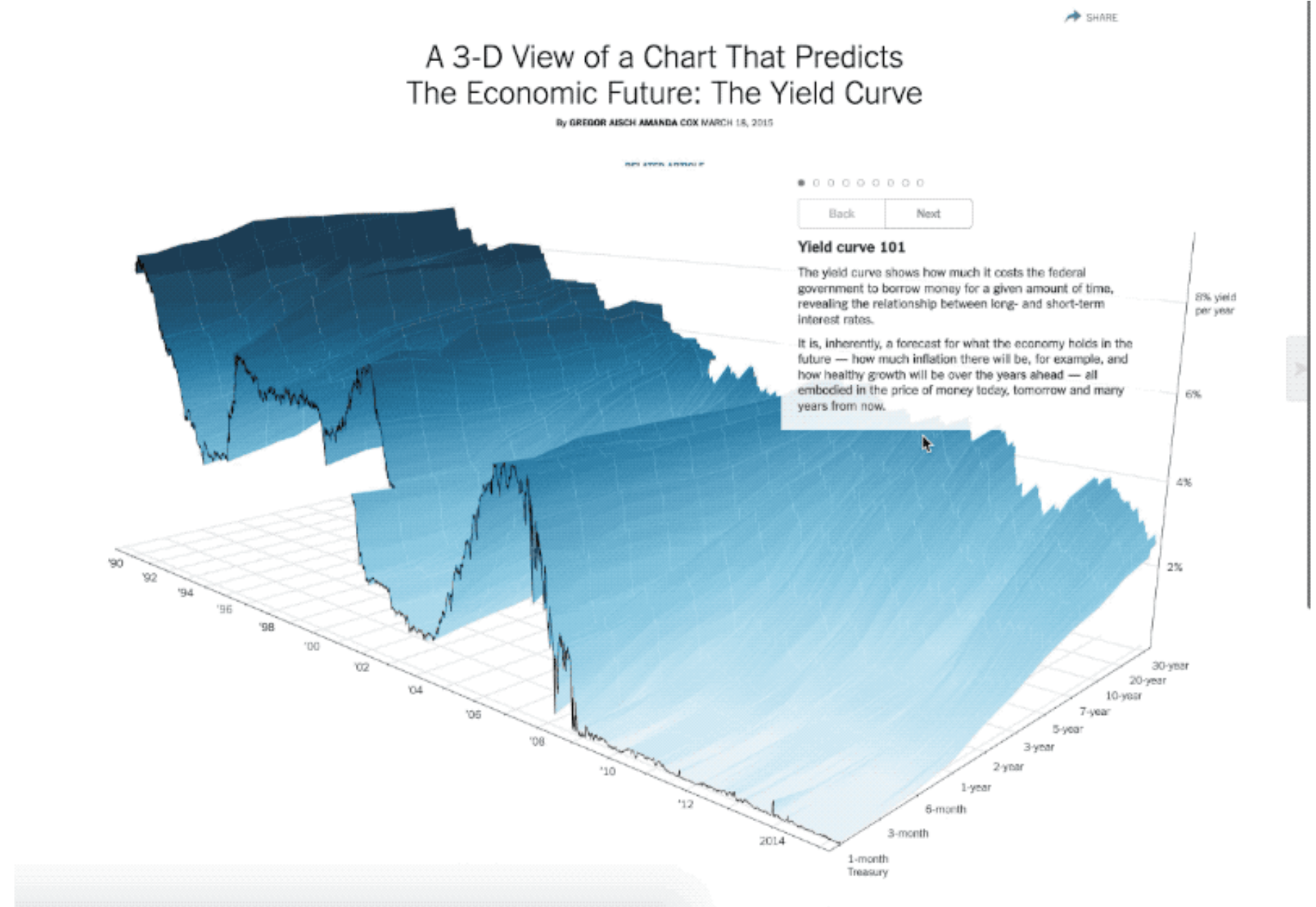
- **Derived data:** cluster hierarchy
- **Juxtapose multiple views:** calendar, superimposed 2D curves
- **Superimposed 2D curves:** The linked 2D views of derived aggregate curves and the calendar allow direct comparison and show more fine-grained patterns.



[\[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.\]](#)

Good Usage of 3D

Constrained navigation
steps through carefully
designed viewpoints.



<http://www.nytimes.com/interactive/2015/03/19/upshot/3d-yield-curve-economic-growth.html>

Example Empirical Evidence

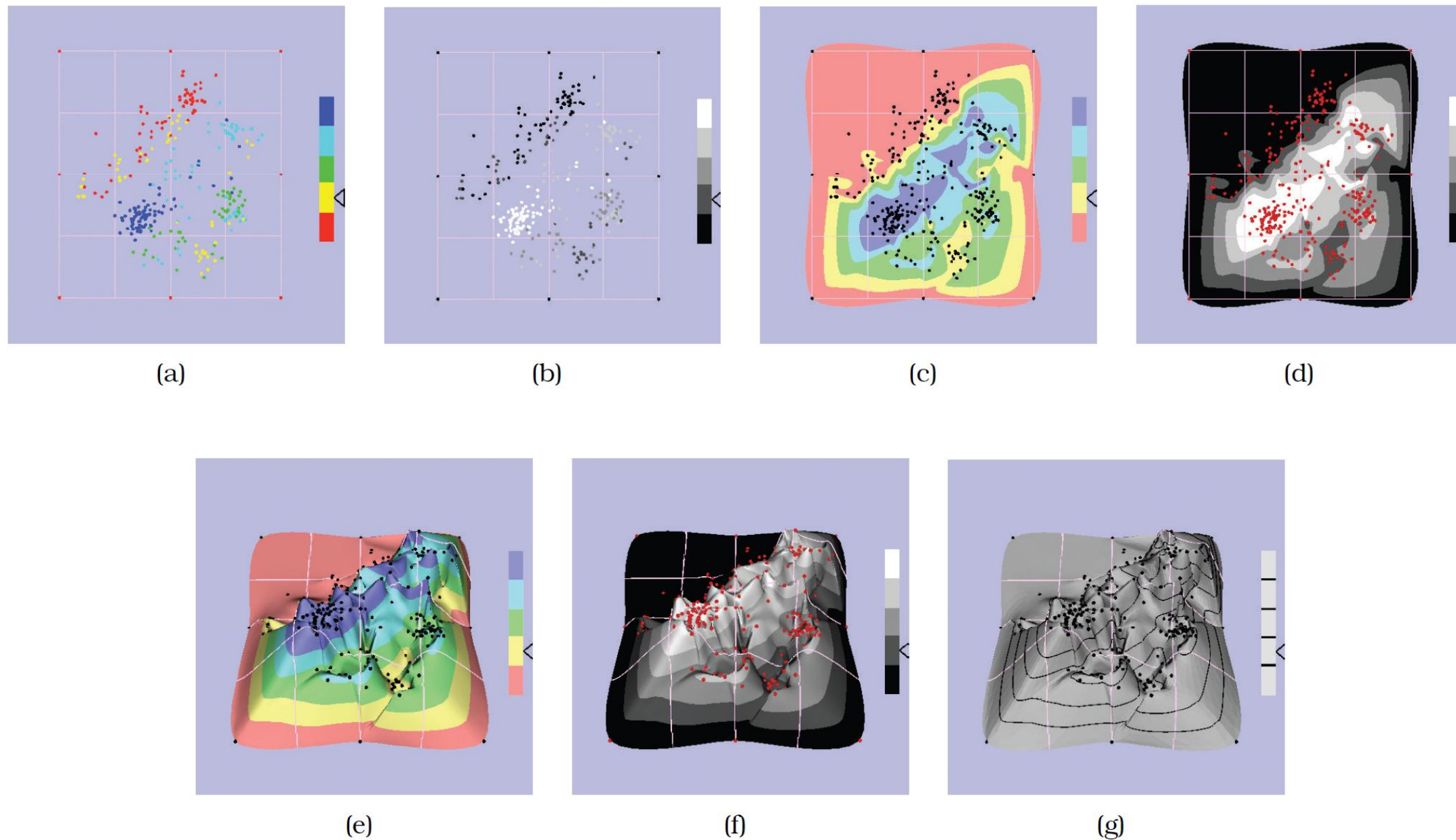


Figure 6.9. Point-based displays were found to outperform information landscapes in an empirical study of visual encodings for dimensionally reduced data. (a) Colored points. (b) Grayscale points. (c) Colored 2D landscape. (d) Grayscale 2D landscape. (e) Colored 3D landscape. (f) Grayscale 3D landscape. (g) Height only. From [Tory et al. 07, Figure 1].

2. No Unjustified 2D

- Consider whether network data requires the 2D spatial layout
 - Especially if reading text is central to the task!
 - Arranging as a network means lower information density and harder label lookup compared to text lists
- Benefits outweigh costs when topological structure/context important for the task
 - be especially careful about search results, document collections, ontologies



➔ Network Data

➔ Topology



➔ Paths



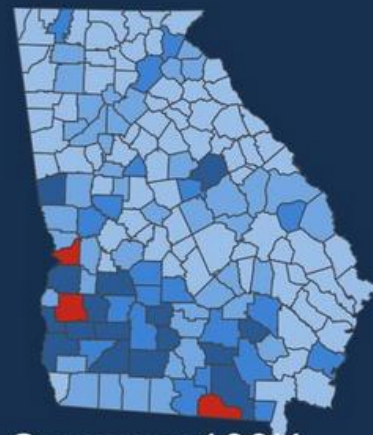
Do you really need a map?



Georgia Person @andishehnourae · Jul 17

In just 15 days the total number of #COVID19 cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/

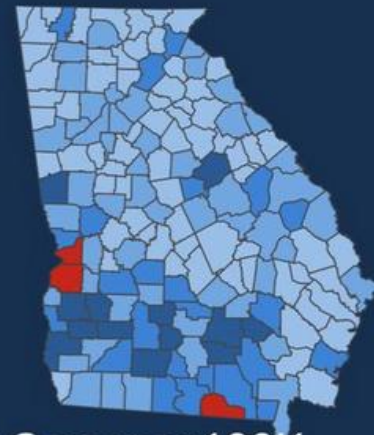
Cases per 100K▼



Cases per 100K



Cases per 100K▼



Cases per 100K



1.7K

26K

47.5K



- They use the position location
- Use them for geographical tasks
- Bad for comparison tasks
- Don't use a map just because you have geo data

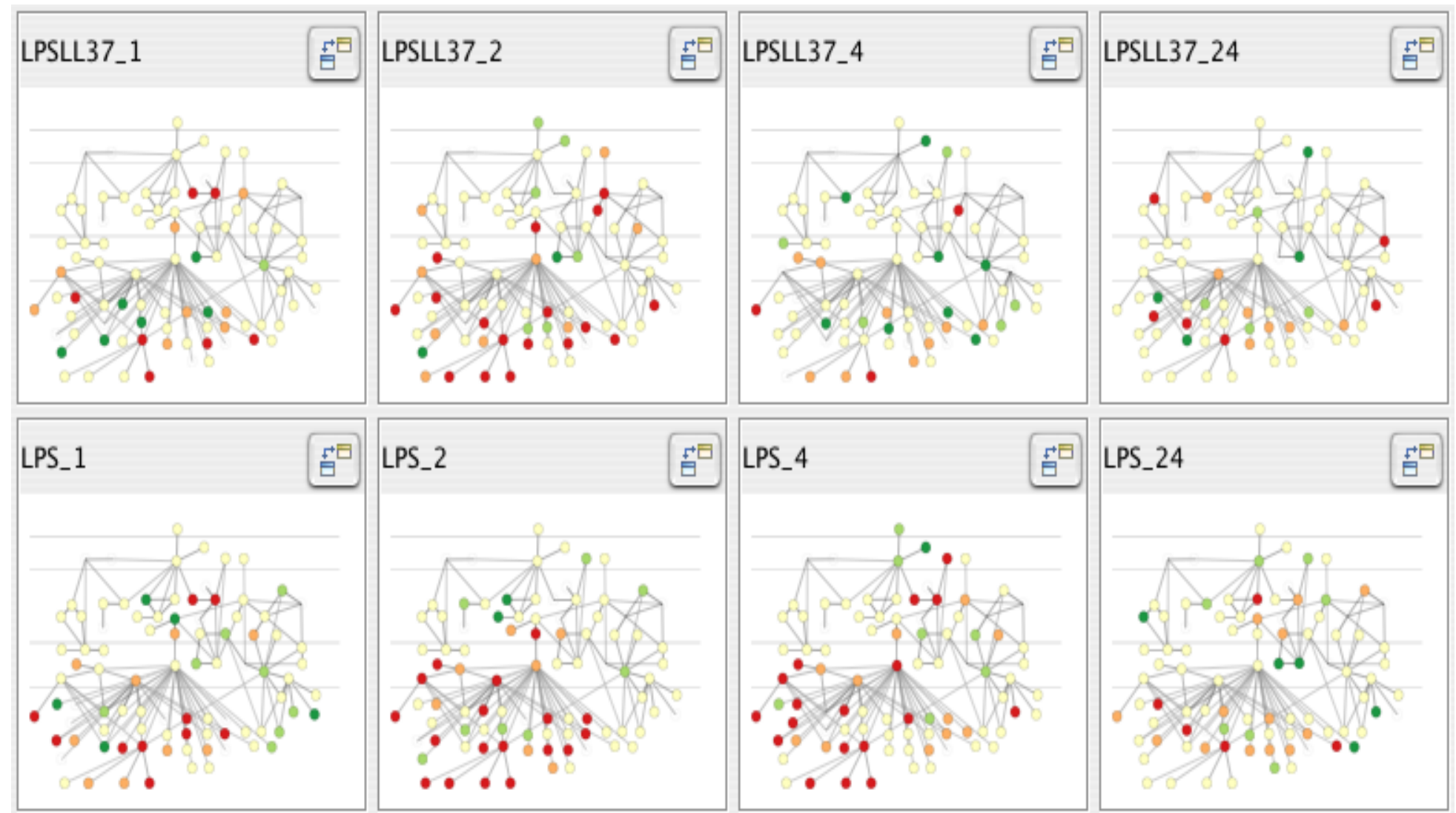
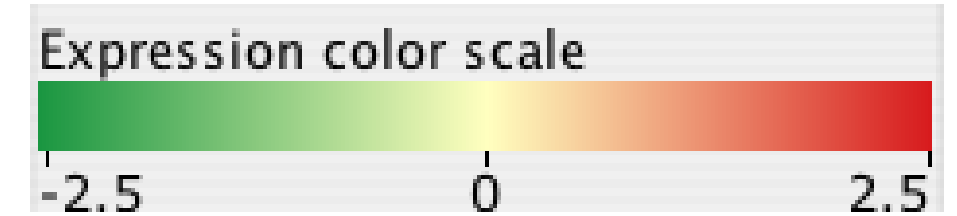
3. Eyes Beat Memory

- Principle: **external cognition VS internal memory**
 - easy to compare by moving eyes between side-by-side views
 - harder to compare visible item to memory of what you saw
- Implications for animation
 - great for choreographed storytelling
 - great for transitions between two states
 - poor for many states with changes everywhere
 - consider small multiples instead



3. Eyes Beat Memory Example: Cerebral

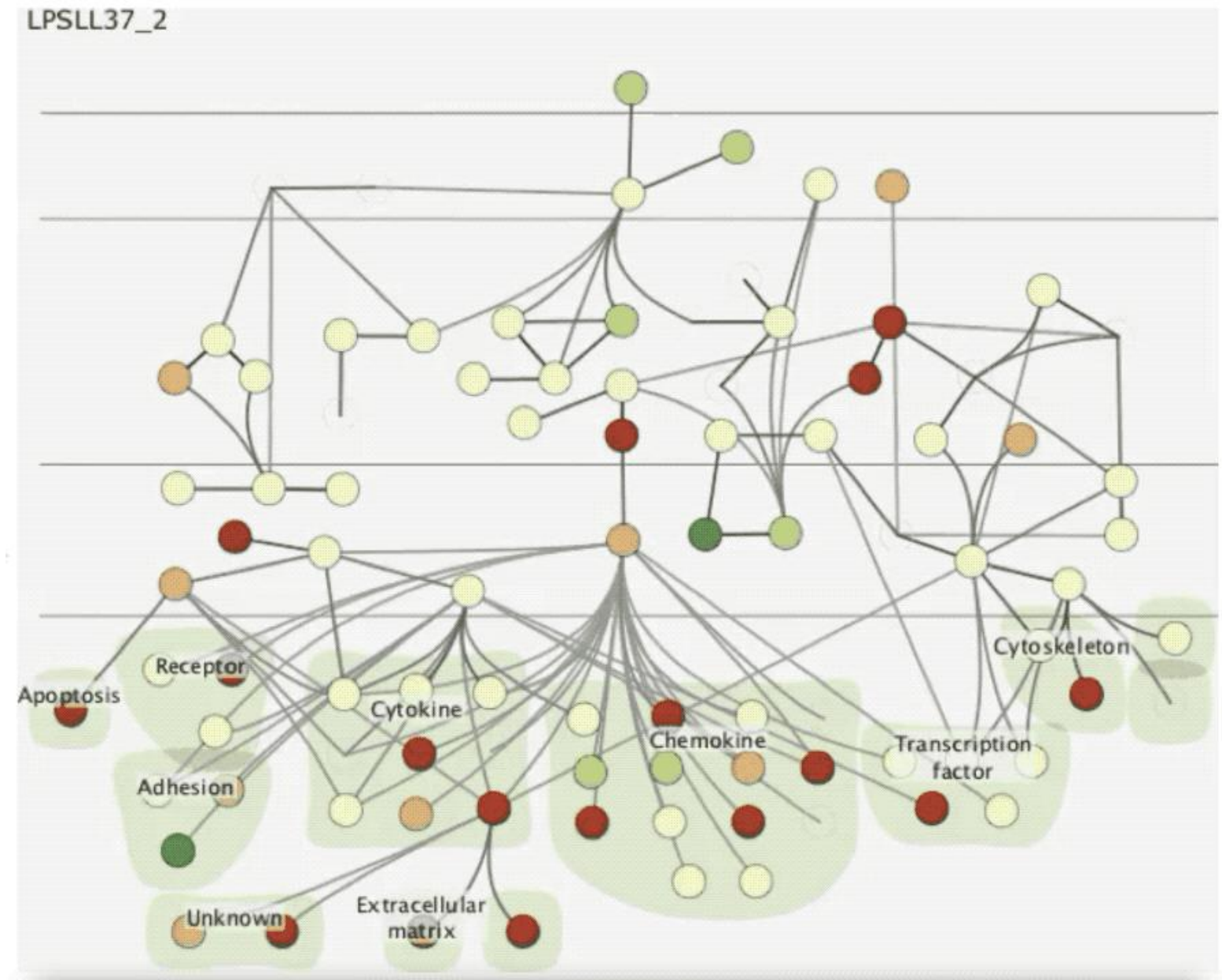
- **Small multiples:** one graph instance per experimental condition
 - Same spatial layout
 - Color differently, by condition



[Cerebral: Visualizing Multiple Experimental Conditions on a Graph with Biological Context. Barsky, Munzner, Gardy, and Kincaid. *IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2008)* 14:6 (2008), 1253–1260.]

Why Not Animation?

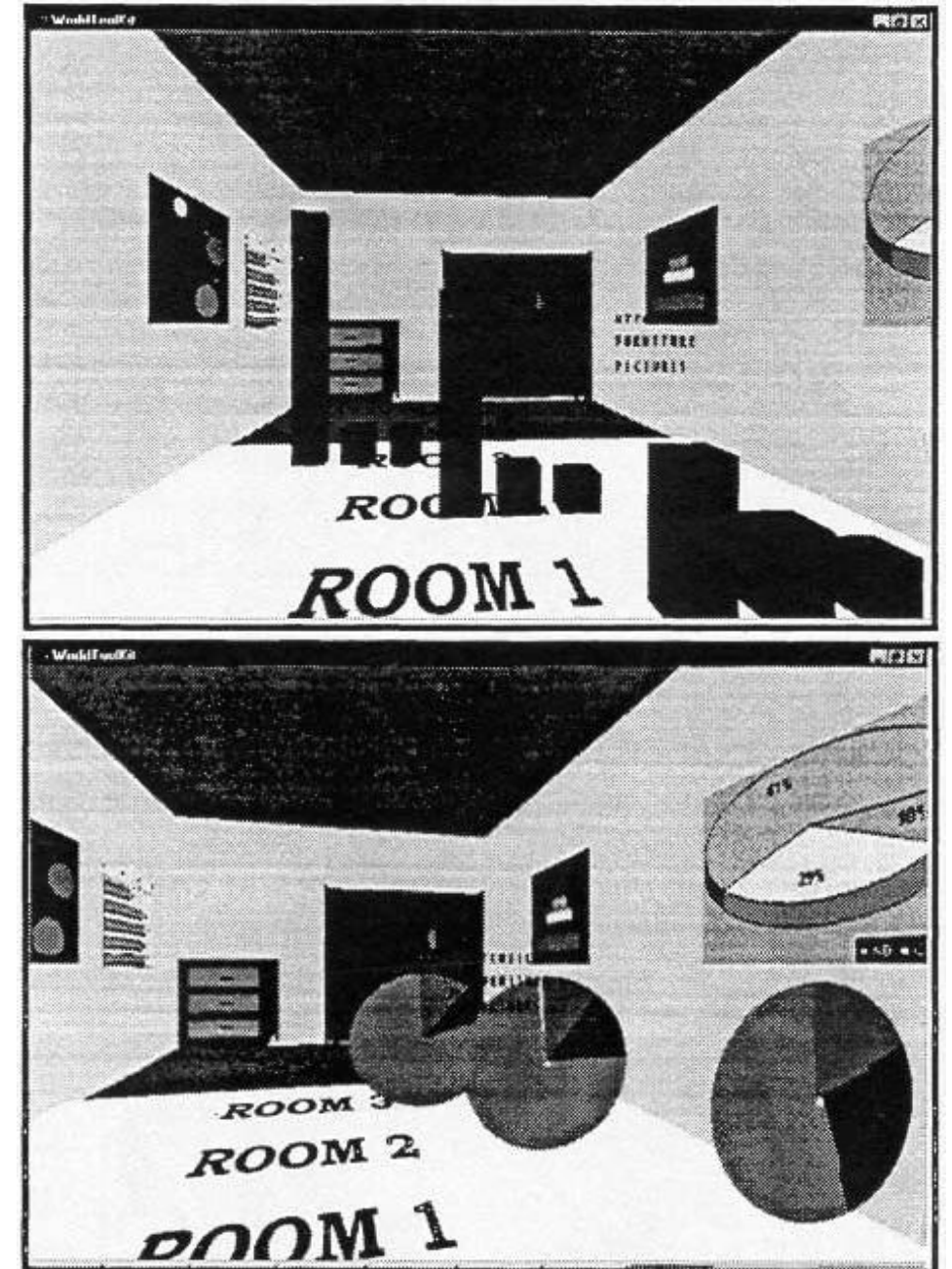
- Disparate frames and regions:
comparison difficult
 - vs contiguous frames
 - vs small region
 - vs coherent motion of the group
- Change blindness
 - even major changes are difficult to notice if the mental buffer wiped
- Safe special case
 - animated transitions



[Cerebral: Visualizing Multiple Experimental Conditions on a Graph with Biological Context. Barsky, Munzner, Gardy, and Kincaid. IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2008) 14:6 (2008), 1253–1260.]

4. Resolution Beats Immersion

- Immersion typically not helpful **for abstract data**
 - do not need sense of presence or stereoscopic 3D
- Resolution much more important
 - pixels are the scarcest resource
 - desktop also better for workflow integration
- **Virtual reality** for abstract data very difficult to justify



[\[Development of an information visualization tool using virtual reality. Kirner and Martins. Proc. Symp. Applied Computing 2000\]](#)

5. Overview First, Zoom and Filter, Details on Demand

- Influential **mantra** from **Shneiderman**

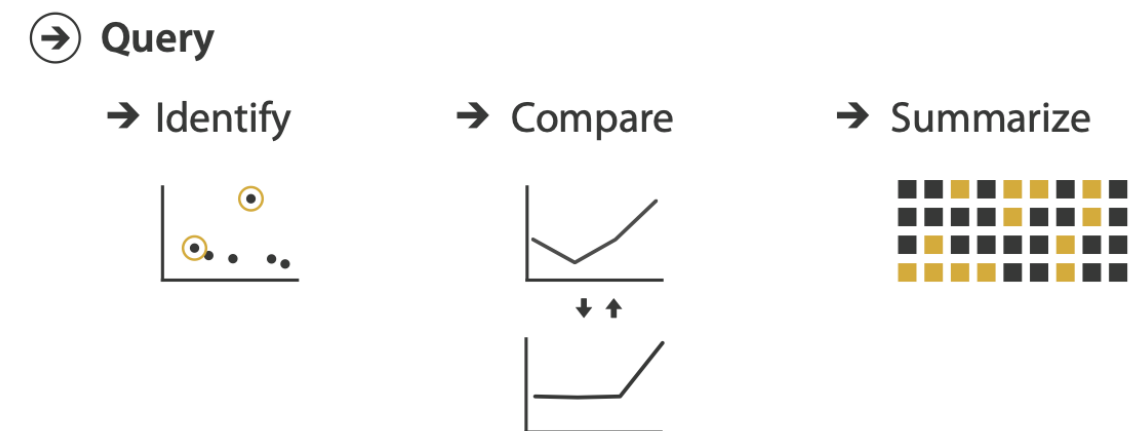
[\[The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. Shneiderman. Proc. IEEE Visual Languages, pp. 336–343, 1996.\]](#)

- Overview = summary

- microcosm of full vis design problem

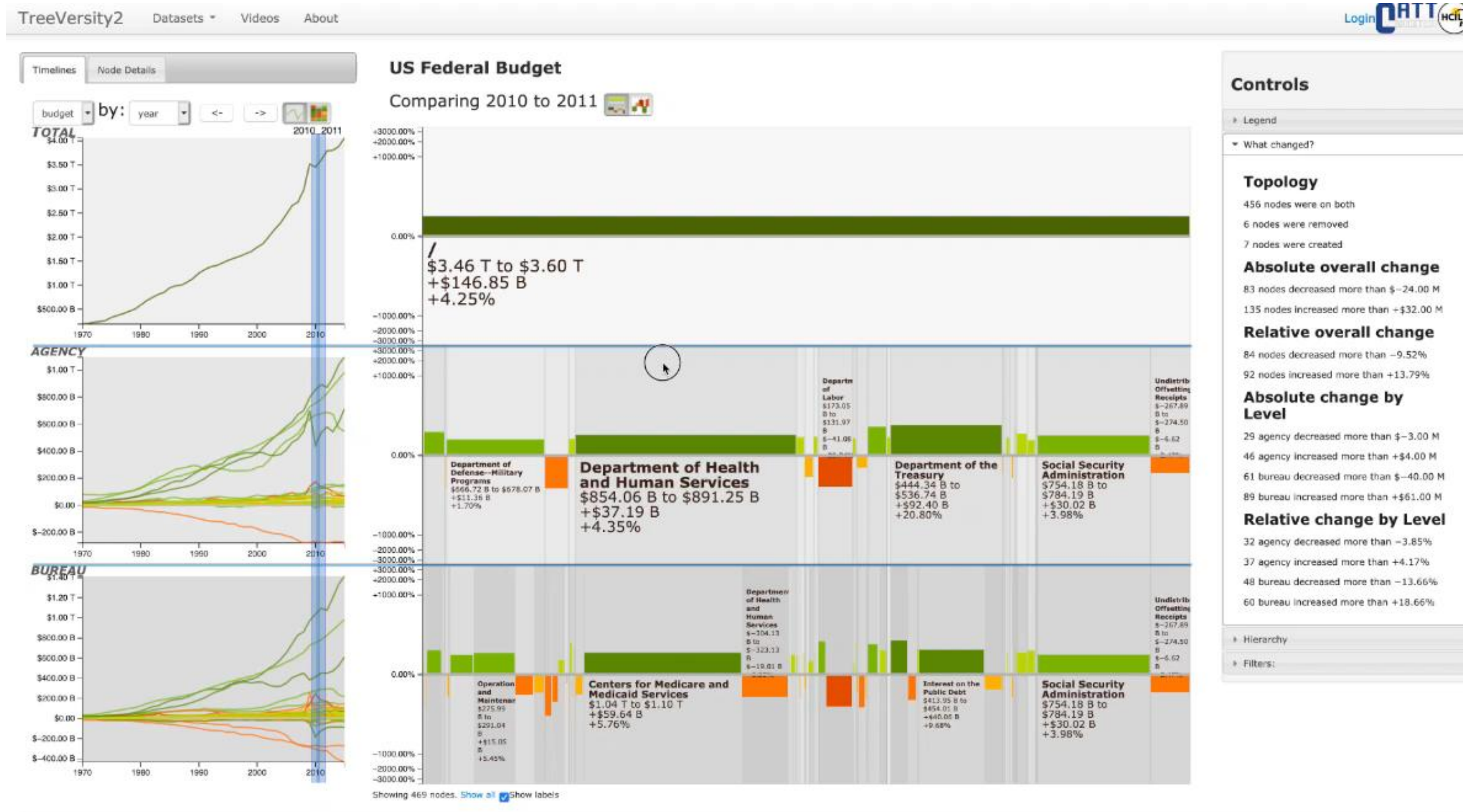
- Nuances

- beyond just two levels: multi-scale structure
- difficult when scale huge: give up on overview and browse local neighborhoods?

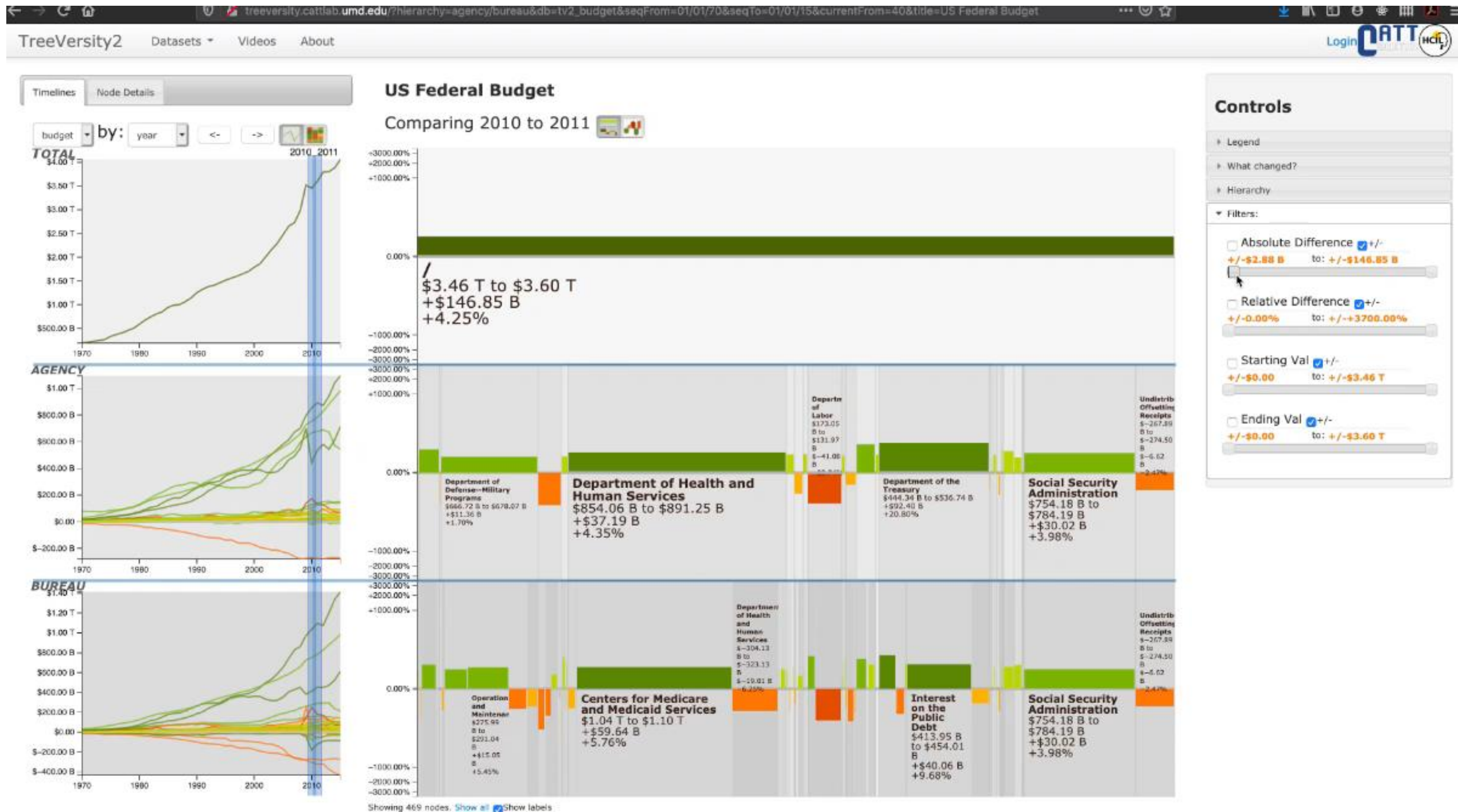


[\[Search, Show Context, Expand on Demand: Supporting Large Graph Exploration with Degree-of-Interest. van Ham and Perer. IEEE Trans. Visualization and Computer Graphics \(Proc. InfoVis 2009\) 15:6 \(2009\), 953–960.\]](#)

Overview → Zoom



Overview → Filtering



6. Response & Interaction

- The **latency** of interaction, namely, how much time it takes for the system to respond to the user action, matters immensely for interaction.
- A system will feel responsive if these are taken into account by providing feedback to the user within the relevant time scale.
- **Perceptual processing** 0.1 sec. = such as screen updates
- **Immediate response** 1 sec. = visual feedback showing what item that user has selected with a mouse click,
- **Brief tasks** 10 sec. = complex tasks into simpler pieces;
- Importance of visual feedback

6. Response & Interaction (responsiveness is required)

- ***visual feedback: three rough categories***
 - *0.1 seconds: perceptual processing*
 - subsecond response for mouseover highlighting - ballistic motion
 - *1 second: immediate response*
 - fast response after mouseclick, button press - Fitts' Law limits on motor control
 - *10 seconds: brief tasks*
 - bounded response after dialog box - mental model of heavyweight operation (file load)
- **scalability considerations**
 - highlight selection without complete redraw of view (graphics frontbuffer)
 - show hourglass for multi-second operations (check for cancel/undo)
 - show progress bar for long operations (process in background thread)
 - rendering speed when item count is large (guaranteed frame rate)

7. Get It Right in Black and White

- The slogan used as a design guideline for effective use of colour
- Ensure the visual representation are **legible** even if the image is transformed from full colour to black and white
- **How?** – check your work by transform it to B & W
- **Suggested – the most important attribute encoded with luminance channel**
 - ensure adequate luminance contrast and considering the **hue** and **saturation** channels as secondary sources of information

8. Function First, Form Next

- Start with **focus on functionality**
 - straightforward to improve aesthetics later on, as refinement
 - if no expertise in-house, find good graphic designer to work with
- **Dangerous to start with aesthetics**
 - usually impossible to add function retroactively

- Guidelines and considerations, not absolute rules
 - when to use 3D? when to use 2D?
 - when to use eyes instead of memory?
 - when does immersion help?
 - when to use overviews?
 - how long is too long?
 - which comes first, form or function?